

	1.8.4	Shape of a Drum	78
	1.8.5	Heat	87
	1.8.6	Relations Between the Two Spectra	91
1.9		Heat and Waves in $\mathbb{E}^3, \mathbb{E}^d$ and on the Sphere ..	94
	1.9.1	Euclidean Spaces	94
	1.9.2	Spheres	95
	1.9.3	Billiards in Higher Dimensions	97
	1.9.4	The Wave Equation Versus the Heat Equation	98

1.1 Preliminaries

Some (not all) of the following generalizes to any dimension; this will be left to the reader. We will only give references for special topics. General references could be Berger and Gostiaux 1988 [175], Coxeter 1989 [409], do Carmo 1976 [451], Klingenberg 1995 [816], Spivak 1979 [1155], Sternberg 1983 [1157], and Stoker 1989 [1160]. For those who like computer programming, Gray 1998 [584] will be of interest. We will assume elementary calculus and also that functions are differentiable as often as needed.

1.2 Distance Geometry

1.2.1 A Basic Formula

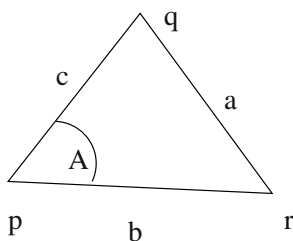


Fig. 1.1. A triangle in $\mathbb{E}^2$

In the Euclidean plane $\mathbb{E}^2$ we have distances between points and angles between smooth curves wherever the curves meet. The equation

$$a^2 = b^2 + c^2 - 2bc \cos A \quad (1.1)$$

should be kept in mind, where a, b, c are the pairwise distances between three points, as in figure 1.1 and A is the angle opposite the side of length a . The triangle inequality in $\mathbb{E}^2$:

$$a \leq b + c$$

is strict, in that equality forces the three points to be colinear, with p between q and r . Equation 1.1 is basic in telemetry, needed when your instant camera measures distances for you. To know the distance between any two of the points, you only need the angle (a local measurement) and two side lengths, not all three.¹ We will encounter other spaces like this later on. Think of mountains in your way along one side, as in figure 1.2. We use equation 1.1 in everyday life, in primitive geodesy, astronomy, automatic focusing, etc.

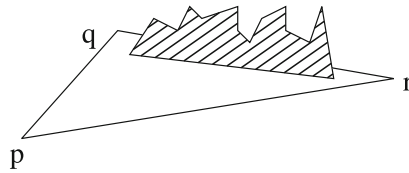


Fig. 1.2. One side is hard to measure

1.2.2 The Length of a Path

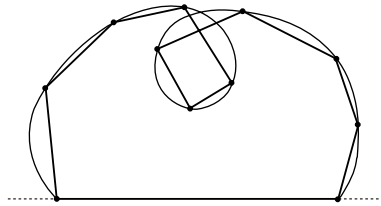


Fig. 1.3. Polygonal approximation

A *path* between two points is a continuous curve between those points. Its *length* is the supremum of the total lengths of polygons inscribed in the curve. The supremum need not be finite, e.g. a fractal curve like a snowflake. But if the curve, $c(t)$, is smooth enough² say with derivative $c'(t)$, then it has finite length $\int \|c'(t)\|$. The strict triangle inequality above ensures that the shortest path between two points is exactly the straight line interval between them.

For any metric space, one can define the length of a path as a supremum as above, and then consider the infimum of path lengths between two points.

¹ However, keep in mind the difficulty of making local measurements; angles between little pieces of straight lines are very sensitive to mismeasurement.

² More precisely, rectifiable; see Wheeden & Zygmund 1977 [1258].

If all points are connected by finite length paths, then this infimum is another metric on the same metric space. In general, it is a different metric, as in the case of the sphere in $\mathbb{E}^3$. The new metric is called the *inner metric*, and if the initial metric coincides with the inner metric, then the space is called a *length space*. Length spaces are becoming more important today. For example, see Burago, Gromov and Perelman 1992 [282] and Gromov 1999 [633].

1.2.3 The First Variation Formula and Application to Billiards

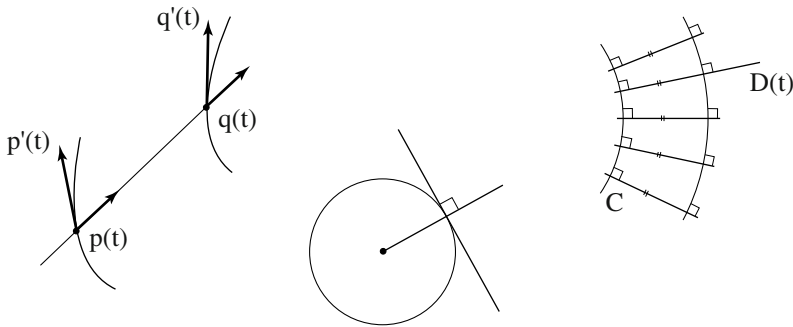


Fig. 1.4. Two points travelling through the plane

Imagine that we have two points $p(t)$ and $q(t)$ running along two curves in the Euclidean plane, as in figure 1.4 and we watch how the distance $d(p(t), q(t))$ varies with time t . The first variation formula is:

$$\frac{d}{dt}d(p(t), q(t)) = (q'(t) - p'(t)) \cdot \frac{q(t) - p(t)}{\|q(t) - p(t)\|} \quad (1.2)$$

Consequently, if we have a one parameter family of straight lines, then any two curves which are perpendicular to the lines must be at constant distance; for example, circles about the origin. So the tangent line to a circle is orthogonal to the radius.

A more elaborate application of this equation, which will be important later on, is the *billiard table*. Suppose that we have a convex domain in the plane, say D , and consider a small particle (think of a small ball, of infinitesimal radius) flying inside D with no force applied to it. The particle will travel on a straight line with constant speed through the interior of D , and when it strikes the boundary ∂D it will bounce and travel on a new straight line. The new straight line path and the old one will be reflections of one another about the normal line of the curve ∂D at the point of impact. We will refer to this as *mirror bouncing*, since we see the same trajectories if we have light rays travelling inside a curved mirror in the shape of ∂D . For the moment, we will

suppose that ∂D is differentiable and strictly convex. We will call our particle a *billiard ball*. The book of Tabachnikov 1995 [1175] is a complete reference on billiards; some other references are Sinai 1990 [1141], Katok et. al. 1986 [788], Arnoux 1988 [70], Veech 1989 & 1991 [1208, 1209], Katok 1987 [785], and Gutkin 1996 [672].

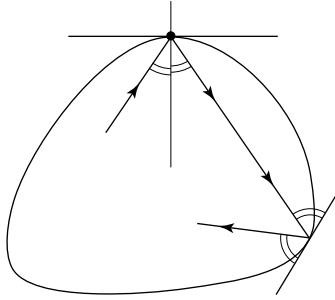


Fig. 1.5. A billiard ball

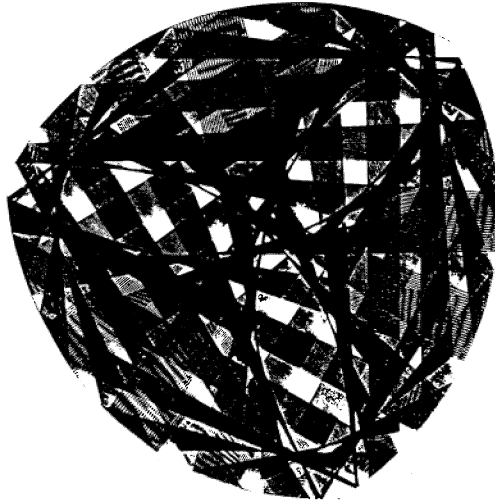


Fig. 1.6. The long term behaviour of the billiard ball

What happens to these trajectories over large periods of time? As we will see in chapter 10, this question is fundamental in rational mechanics and almost every field of physics. It may happen that a trajectory is periodic, returning eventually to its original position and velocity. The search for periodic trajectories is a corner stone in the study of long duration dynamics.

We devise the length counting function to try to count periodic trajectories, setting $CF(L)$ to be the number of periodic trajectories of length less than L . Let us now see how to exhibit some periodic trajectories.

In 1913, G. D. Birkhoff remarked that the first variation formula yields many periodic trajectories. Pick any number P of points of ∂D , in a fixed order (not necessarily consecutive along ∂D) as in figure 1.7, turning around the boundary Q times. Draw the polygon connecting those points in that order. Using the first variation formula, we find that mirror bouncing is exactly the condition that the polygon has critical length under perturbing its vertices along ∂D . But now for each choice of P and Q , we can take the polygon of greatest length, which exists by compactness of the space of polygons with given P and Q values. N.B. this construction does not yield all of the periodic trajectories. We will return to billiards in §§1.8.6.

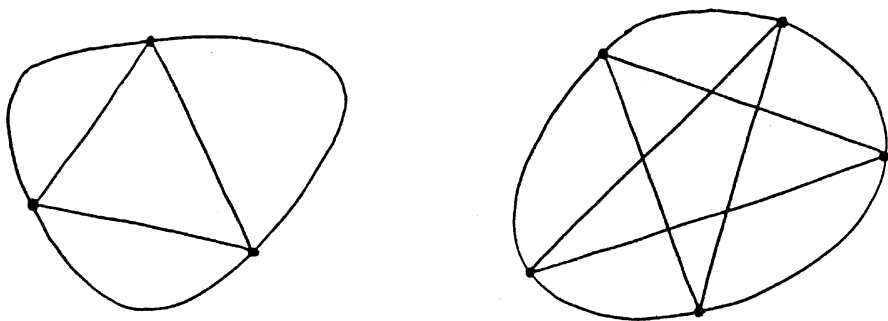


Fig. 1.7. Two of Birkhoff's configurations

Let us take our domain D to be a convex polygon, a case we expect to be simpler than a general convex curve. We shall be careful to eliminate trajectories which strike a vertex of the boundary. It is still an open question if there is even one periodic trajectory. This question is solved for polygons with angles rational multiples of π (see below), but is open for general polygons, even triangles. If the triangle is not obtuse, I leave it to you to find a (classical) periodic trajectory (hoping that you are not obtuse either). Everyone expects that infinitely many periodic trajectories exist, as for rational angled polygons and convex domains.

Rational angled polygons and convex domains are the only domains for which periodic billiard trajectories are well understood. To the rational angled polygons, one can associate a compact space, a generalized Riemannian manifold, locally Euclidean but for $2\pi k$ singularities, k an integer. See Arnoux 1988 [70] for more information. One knows that $CF(L)$ is asymptotically quadratic in L , as $L \rightarrow \infty$. See Vorobets, Gal'perin, and Stëpin 1992 [1224] and Gal'perin and Chernov 1991 [545].

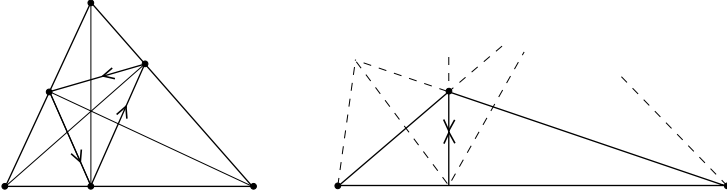


Fig. 1.8. Two triangular billiard tables

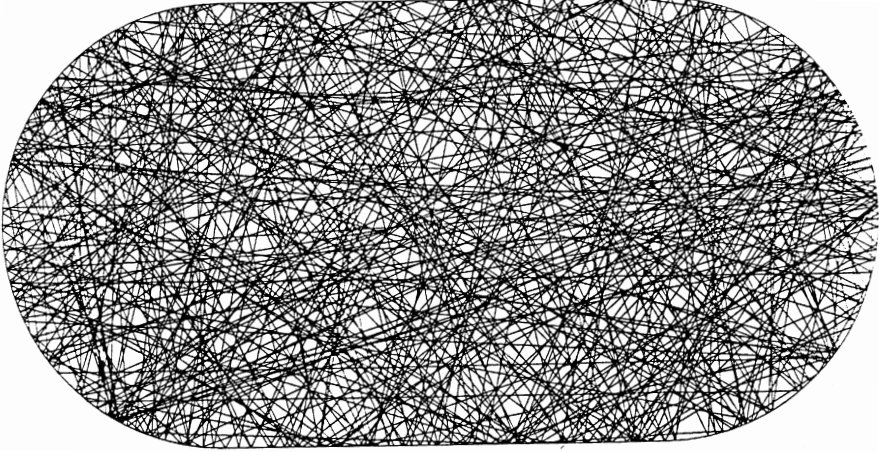


Fig. 1.9. The Bunimovich Stadium

In concave domains, beams of light or pencils of trajectories are dispersed when reflected from the boundary, as in figure 1.10 on the following page and this implies *ergodicity* (see §§§10.5.1.1 for the definition). See Katok et. al. 1986 [788] for the proof. Finding a simple proof is an open problem. However, a good deal is known about concave domains; for example let us just mention that $CF(L)$ is asymptotically exponential in L . Also see §§1.8.6 below for a heuristic relation between concave billiard tables, statistical mechanics, and manifolds with negative curvature.

For both rational angled polygons and convex domains, rigorous proofs of existence of periodic trajectories are quite difficult. There is a third case, a generalization of the *Bunimovich stadium* which is not completely mysterious; see section 5.3 of Tabachnikov 1995 [1175] for a modern proof that there are periodic trajectories, and also see the references therein. In this case we have ergodicity and $CF(L)$ is exponential in L . These examples are convex (but with flat parts) and came as a shock to the ergodic community in 1979 when most people were convinced that some concavity was required for ergodicity, or if you prefer, that convex domains had too much focusing to achieve ergodicity.

Defocusing is apparent in figure 1.10. In §§1.4.3 we will see an interesting curvature condition needed to construct these examples.

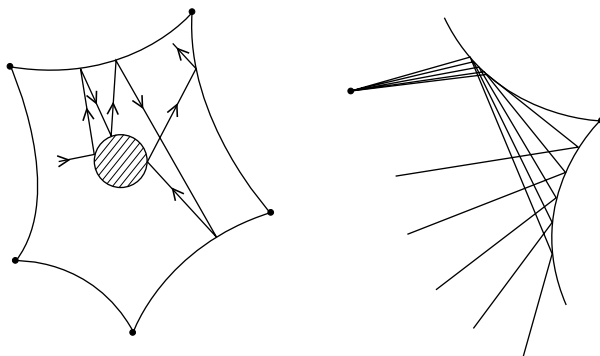


Fig. 1.10. Spreading rays in a nonconvex billiard table

A strictly convex billiard table can not be ergodic, because its flow has caustics (at least if its boundary is six times differentiable, or perhaps less—the optimal result is open), by a fundamental result of Lazutkin, discussed on page 92. But it is an open problem whether there are smooth ergodic billiard tables (the Bunimovitch and Wojtkowski examples are not smooth, having flat parts joined to curved parts with nonvanishing curvature).

One does not know how general these examples are. Let us mention two general results. The first, due to Katok, asserts that for polygonal billiards, $CF(L)$ grows more slowly than exponentially. See Katok's 1987 [785] and also the nice survey of Gutkin 1996 [672]. The proof is extremely interesting and consists of two parts. The first part studies the trajectories using successive reflexions on the sides, unwrapping the trajectory to make it a straight line on the plane. But this is not enough: the second part resorts to a strong theorem in ergodic theory connecting topological and metric entropy. We will define entropy in §§§10.5.1.2.

The second result is a density theorem in the set of all convex billiard tables, found in Arnoux 1988 [70]. For concave billiard tables, the length counting function $CF(L)$ is very well understood. One has the precise asymptotic expression

$$CF(L) \sim \frac{\exp hL}{hL}$$

as $L \rightarrow \infty$ with h the entropy, which is known to be positive. For information on higher dimensional billiards, see §§ 1.9.3 on page 97.

1.3 Geometry of Plane Curves and Two Dimensional Point Kinematics

1.3.1 Length

We will briefly recall some classical observations concerning plane curves, but also some less classical ones. Our style will probably appear different from standard expositions. This is intentional, in order to introduce the viewpoint of Riemannian geometry. So even if you think you know everything about plane curves and point kinematics, you might do well to glance at what follows.

We have a plane curve c with a point $c(t)$ running along it, a function of a parameter t , as smooth as needed (i.e. differentiable as many times as needed). The velocity of this curve c is $c'(t)$. As we have already noted, the length of c from $t = a$ to $t = b$ is

$$\int_a^b \|c'(t)\| dt .$$

This gives a distance on the curve, which is the time needed to traverse from $c(a)$ to $c(b)$ with unit speed. This inner geometry is the same as the geometry of an interval. By *inner* we mean that you care only about what happens on the curve itself, and not what happens outside. Consequently, all curves having the same length are *isometric*, namely they are the same as metric spaces. Equivalently, there is a one to one map which is an *isometry* between them (i.e. preserves distances). Moreover all of them are isometric to an interval of the same length on the Euclidean line $\mathbb{E}^1$. Surfaces will be completely different; see §§1.6.1.

Alternately, look at the triangle inequality. For the inner metric of a plane curve, at least locally,

$$d(p, r) = d(p, q) + d(q, r)$$

for any three points p, q, r with q between p and r . On a circle, between is a local notion, as you see in figure 1.11. Or think about the word “between”. If we call a curve *closed* when it is periodic, then such a curve has a total length, and two closed curves are isometric exactly when they have the same length.

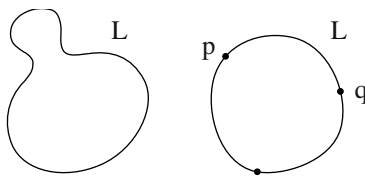


Fig. 1.11. These two curves are isometric

An important remark is in order: the attentive reader might have been puzzled, even outraged, that we never distinguished between a curve as a map $t \rightarrow c(t)$ from an interval of the real line, and its image as the set of points $c(t)$ when t ranges through the values of that interval. It is not easy to have a clear view of these two notions, and the relations between them; they are obscure even in many books. We refer the interested reader to chapter 8 of Berger and Gostiaux 1988 [175] for a very detailed exposition of this point. Now we will just say a few words about this distinction.

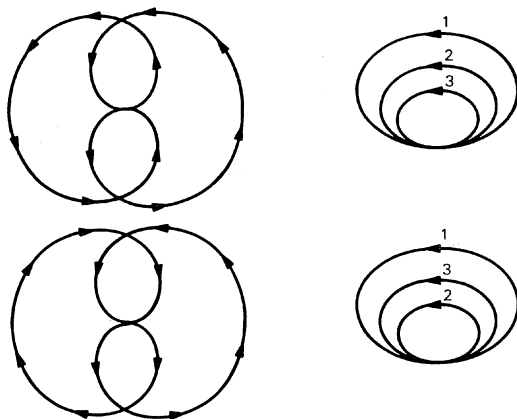


Fig. 1.12. Geometric images you can travel through on different routes

Maps $t \mapsto c(t)$ from an interval of $\mathbb{E}^1$ into $\mathbb{E}^2$ should be called *parameterized curves*. Most often only *regular curves* are allowed, i.e. the speed $c'(t)$ is never zero. Note that here, and also in the sequel, we use two different notions of how quickly something moves: the *speed* $\|c'(t)\|$ which is the norm of the *velocity* $c'(t)$ (see Feynman, Leighton and Sands 1963 [516], page 9-2). We skip intermediate developments (again, see Berger and Gostiaux 1988 [175]) and jump directly to the concept of a *geometric curve*: a line in the plane (not necessarily a straight line). By this we mean a subset of the plane which, near any point, can be reshaped into a straight line, i.e. taken by diffeomorphism of an ambient region of the plane into a straight line. In modern mathematical jargon, such an object is called a *one dimensional submanifold of the plane*. The link between the two notions is that the image set made by all of the $c(t)$ is always, but in general *only locally*, a geometric curve, and conversely a geometric curve admits regular parameterizations, in particular by arc length (i.e. with constant unit scalar speed). In the sequel, we will most often leave it to the reader to understand if parameterized or geometric curves are considered.

A word about orientations of curves: parameterized curves are automatically oriented, while geometric curves can be given two orientations. When you change parameterizations of a curve, two invariants appear: the first is

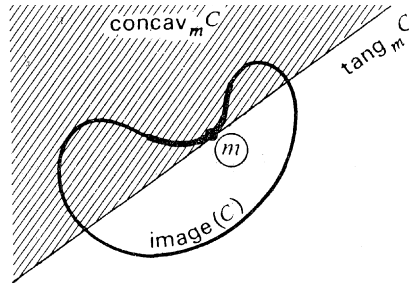


Fig. 1.13. Concavity and tangent line

the *tangent line* to which every $c'(t)$ belongs. The second is defined only for *biregular* curves, i.e. curves for which the acceleration $c''(t)$ is not proportional to the velocity $c'(t)$. The acceleration will then remain in the same open half plane under any reparameterization, and this half plane is called the *concavity* of the curve. It does not depend on orientation. If you parameterize the curve by arc length, the two orientations yield two opposite unit velocity vectors, but one acceleration vector, which is then attached canonically to the curve.

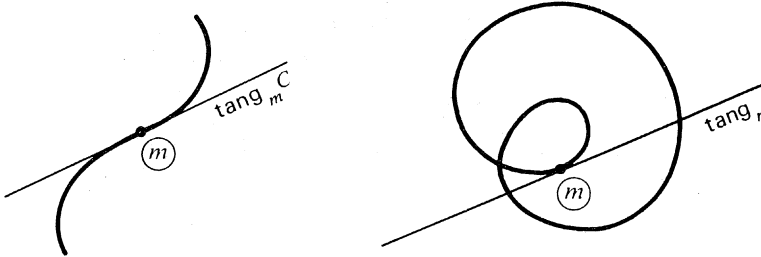


Fig. 1.14. Tangent lines

Everything said here is valid with obvious modification for curves in any Euclidean space $\mathbb{E}^d$.

A closed (periodic) plane curve is said to be *simple* if it is a one-to-one map up to the period (this for a parameterized curve). For a geometric curve, *simple* means a curve which has the global topology of a circle (in the jargon, it is a differentiable embedding in the plane of the circle seen as an abstract one dimensional manifold). We also will assume the speed never vanishes, indeed that it has unit speed.

Do not think that everything is known today about plane curves. A famous problem is to prove that in any plane closed curve, one can inscribe a square. For this problem, and other open ones, see the book Croft, Falconer & Guy 1994 [410]. The oldest unsolved problem in plane geometry is the *equichordal problem*, from 1916: prove that no plane curve can have two equichordal points,

i.e. points so that all chords through them have constant length. The problem is studied extensively in note 6.3, chapter 6 of Gardner 1995 [548]. Do not miss the recent revolution in the theory of plane curves: see the end of §1.4.

1.3.2 Curvature

We now look at the outer geometry of plane curves. The inner geometry of curves does not differ from that of straight lines, but the geometry is radically different if we look at the way a curve sits in the plane. We are going to introduce a concept of *curvature* which measures how much a curve differs from a straight line. For a curve seen as a kinematic motion, curvature is directly linked with the *acceleration* vector $c''(t)$. For a geometric curve, we can introduce it by looking at the variation of the length of the curves drawn at a constant distance, called the *parallel curves* (recall the first variation equation 1.2 on page 4). If the absolute value of the infinitesimal change of length of these equidistant curves, close to a point $m = c(t)$, is the same as for a circle of radius r we say that the curve c has *radius of curvature* r at $m = c(t)$, and that its curvature is $K = 1/r$. The formula to compute curvature is

$$K(t) = \frac{1}{r(t)} = \frac{|\det(c'(t), c''(t))|}{\|c'(t)\|^3}$$

where t is any parameter, and boils down to

$$K(t) = \frac{1}{r(t)} = \|c''(t)\|$$

when t is a unit speed parameterization (i.e. arc length parameterization). Of course a circle of radius R has constant curvature equal to $1/R$. Another way to look at it is the following: the circle $C(t)$ which is defined as tangent to the curve at the point $c(t)$ and has radius $r(t)$ is the circle which has the most intimate contact with the curve (technically, the contact is third order, meaning that the curve and the circle have the same first three derivatives at that point). This circle is called the *osculating circle* to the curve at $m = c(t)$.

To rest a little, we mention an interesting fact rarely found in textbooks. When at a point m of the curve c , the curvature varies, that is to say $dK/dt(m) \neq 0$, then the osculating circle at m crosses the curve (guess where it is inside, and where outside). A baffling consequence is that the osculating circles of a curve whose curvature is never critical never intersect one another. See the picture in figure 1.15. This picture gives two counterexamples. First, consider the unit tangent vectors to this family of circles. We get a continuous vector field, alias a differential equation in the plane. Now at every point of the curve there are at least *two* different integral curves of that vector field: namely the curve and the circle. The moral is that uniqueness of solutions of ordinary differential equations does not follow only from continuity. Second, consider the envelope of this family of circles. In many books it

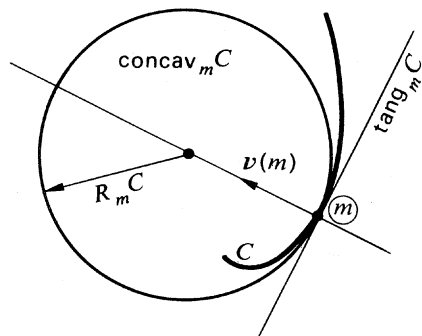


Fig. 1.15. Finding tangent line, concavity and curvature

is “proven” that the envelope is the limit of the so-called characteristic point, the point of intersection of close-by curves. This is wrong (unless we invoke “complex circles”).

There is another geometric definition of the curvature. Parameterize the curve by arc length, so that $\|c'(t)\| = 1$ identically. Then $c'(t)$ is a unit vector, running through the unit circle of the plane. The curvature is the speed of its rotation. More precisely, one needs first to orient the plane, and choose an oriented direction in the plane. Then one can define an angle $\alpha(t)$ for $c'(t)$ which is a real number, defined only up to integer multiples of 2π . The curvature is $K(t) = d\alpha/dt$.

To get rid of the 2π ambiguity, there is an important fact which we are going to use below. For regular curves, when t runs through the interval of definition, one can follow $\alpha(t)$ by continuity and then define a map $t \mapsto \alpha(t)$ into $\mathbb{R}$. Note that if the curve is closed with period T then $\alpha(t + T)$ will in general differ from $\alpha(t)$ by an integral multiple of 2π .

We saw above that straight lines are the shortest paths between two points. Let us look at this again, now in a more sophisticated way. We start with a curve c with ends $p = c(a)$ and $q = c(b)$. Suppose it is as short as any curve can be with these extremities. Consider any one parameter family of curves c_μ with the same ends, with $c = c_0$, and compute their lengths. The fact that the length should be a minimum implies in particular that the first derivative of the length as a function of the parameter μ has to vanish. Computation yields

$$\left. \frac{d \text{length}(c_\mu)}{d\mu} \right|_{\mu=0} = - \int_a^b \langle f(s), c''(s) \rangle ds \quad (1.3)$$

where the vector valued function $f(s)$ (which we can choose to be orthogonal to the curve everywhere) estimates the derivative of the normal displacement of the family of curves. This is also a first variation formula. Note that this formula is nothing but the integrated version of the definition we previously gave for curvature. A classical calculus trick (which is not obvious) proves that if we want all of these derivatives to vanish, namely the above integral

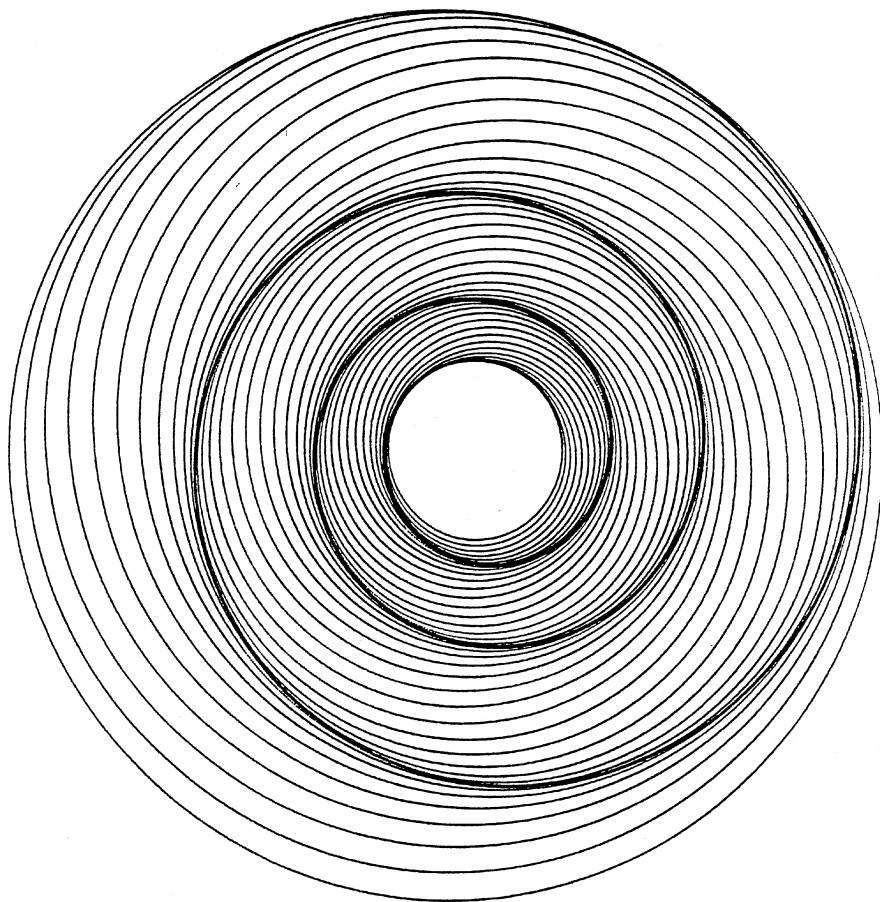


Fig. 1.16. Picture drawn for us by Etienne Ghys

to be 0 whatever the function f vanishing at $s = a$ and $s = b$, then $c''(t) = 0$ identically, and in particular our curve should have everywhere vanishing curvature. Curves with everywhere vanishing curvature are, of course, straight lines. Beware: what we have proven up to now by this method is that shortest paths can only be found among straight lines. Not every zero of the derivative of a function is a minimum. For a local minimum, a sufficient condition is to have positive second derivative. But we still have to search for an absolute minimum. The above proof is interesting because it is the simplest model of a field called *the calculus of variations*, and gives a general scheme to start searching for shortest paths. We will use it soon in §1.6 but principally in chapter 6. To conclude this story for the Euclidean plane, we know that if shortest paths exist, they should be straight lines. Then we check directly as above.

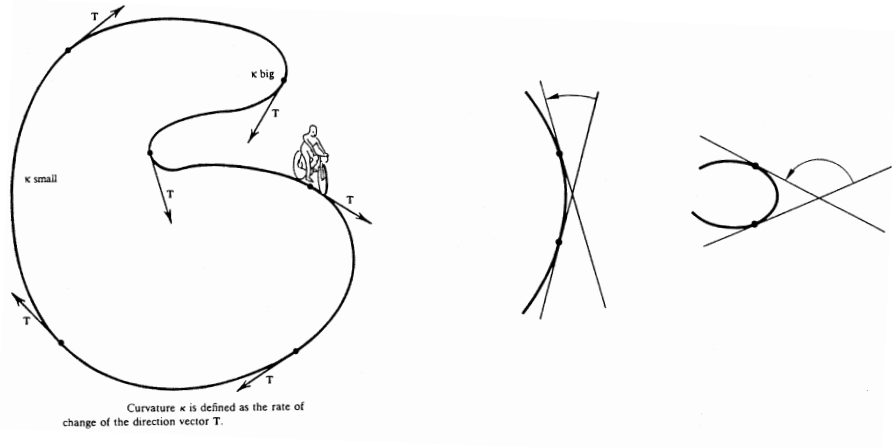


Fig. 1.17. Different definitions of curvature

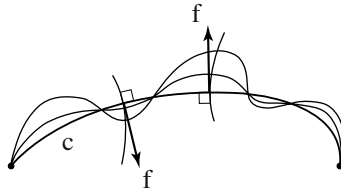


Fig. 1.18. Planar calculus of variations

There is a notion of *algebraic curvature* k as opposed to the scalar curvature K , linked by $K = |k|$. This is defined as soon as the curve is regular and oriented and the plane is also oriented, or (equivalently) if one side of the geometric curve is chosen, or (equivalently) if one of the two possible unit normal vectors is chosen. This side need not be the concavity side. Denote by $n(t)$ the unit normal vector you have chosen. Remember now that $c''(t)$ is an invariant. The algebraic curvature of c at t is the real number $k(t)$ so that

$$c''(t) = k(t)n(t)$$

A nice theorem about algebraic curvature:

Theorem 1 (More tightly curved curves) *If two curves c_1, c_2 are arc length parameterized, and have respective curvatures satisfying $k_1(t) \geq k_2(t)$ and start at the same initial point in the same direction, then the curve c_1 lies entirely inside the curve c_2 .*

(because it is “more curved” as intuition tells you). For a proof see Chern 1989 [366] and exercise 7, section 5-7 of do Carmo 1976 [451]. Beware that you need simple curves if you want a global result. If not, you just get a local one. The

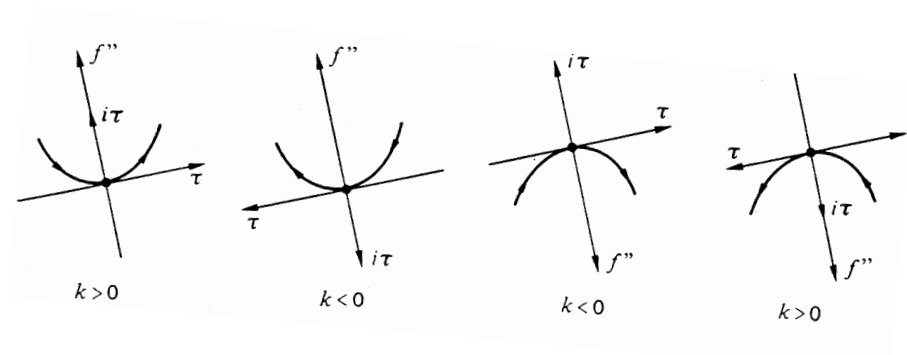


Fig. 1.19. Picturing algebraic curvatures

above theorem is one of the simplest ones in *control theory*. You will see many, many others later on. Note the following amusing consequences. Consider a simple closed plane curve, and the point where the curvature is maximum. Then the osculating circle at that point can keep rolling inside the curve all the way around. Similarly, the curve itself can roll inside its osculating circle of minimum curvature.

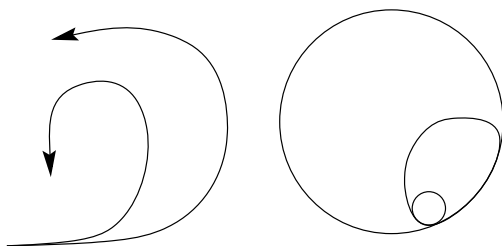


Fig. 1.20. (a) One is more curved than the other (b) The little circle can roll inside the closed curve, and the closed curve can roll inside the big circle

Let us make more precise our claim above that curvature is the only invariant of plane curves. More precisely the algebraic curvature is a characteristic invariant: if in the above inequality for curvatures we have equality ($k_1(t) = k_2(t)$ for every t) then the curves c_1 and c_2 are the same. Moral: if you know the value of the curvature as a function of the arc length then you know the curve completely. If the curves do not have the same initial point and direction, this just means that they are deduced one from the other by a Euclidean motion. The simplest example: a curve with non-zero constant curvature is part (or the whole) of a circle.

A spectacular application of this property of arcs of circle is the manufacture of a perfect straightedge. Take two pieces (of anything suitable for the job) and rub one against the other. The above theorem tells you that such a rubbing will eventually carve the two bounding curves into perfect circles. Now take three pieces, and rub them two by two. The boundaries will be circles, but the pieces will have to exhibit opposite convexity, i.e. equal algebraic curvature with opposite sign. For three, this is not possible unless the common curvature is zero. Naively, one might imagine that a laser can draw straight lines, but this is impractical. Today, lasers are used only to check the accuracy of the straightedge, after it is manufactured following the above recipe; once the laser has reported its evaluation of the straightedge, corrections are carried out by skilled hands using emery powder. Before the invention of the laser, accuracy was tested with interferences.

The same technique is still the only one available to make “perfect” planes: rub three surfaces against one another, and the result has to be a plane. To prove it, use the fact which we will quote on page 47 that pieces of a surface, all of whose points are umbilics, are pieces of spheres. To check accuracy, one doesn’t use a laser, but drags a straightedge around on the surface, or a comparison plane, or applies an interferometer.³ Rubbing only two pieces will yield pieces of a sphere, and is still the only method to sculpt spherical lenses.

In the same spirit, to fashion “perfect” balls, (which have thousands of practical applications, for example as ball bearings), one method was to put approximate balls inside a box, and stir them. Proof that the limit object is a sphere is not as elementary as proofs of the efficacy of the previous production methods, and was only recently established in Andrews 1999 [52]. Since 1907, balls for ball bearings have been produced by rolling rough balls between two plates with toroidal concentric grooves and random indentations. See Berger 2003 [173] or technical books for more details.

For the story of curves in $\mathbb{E}^3$ see Berger and Gostiaux 1988 [175], do Carmo 1976 [451], or Stoker 1989 [1160]. We will just mention briefly what happens there. This time there are two characteristic invariants, curvature and torsion, but the curves need to be biregular, i.e. the acceleration should never vanish. Things also extend similarly to any $\mathbb{E}^d$, this time with $d - 1$ invariants: see Spivak 1979 [1155]. But the first invariant is always the curvature and is defined simply as the norm $\|c''(t)\|$ for any arc-length parameterization. Only straight lines have everywhere zero curvature. For the kinematician this is the old fact that points with no force applied to them move along straight lines and at constant speed.

Take helices for example. They can be characterized as the curves with constant curvature and torsion. A practical application, in the spirit of the one just above: graduate a straightedge to make ruler; this is equivalent to

³ An *interferometer* is a device which uses the interference of two waves (radio, acoustic, or light waves will do) to make very precise distance measurements.

making a protractor, dividing a circle in equal parts, etc. It is clear that you can do it if you can produce a screw (a helix). This is even simpler than a straightedge. Just start with an approximately good pair of a bolt and a nut and rub one against the other. At the end we will get a perfect bolt, and a perfect nut, because the common boundary will be made up of a union of curves with necessarily constant metric invariants, hence easily seen to be helices.

1.4 Global Theory of Closed Plane Curves

1.4.1 “Obvious” Truths About Curves Which are Hard to Prove

Systematic references for what follows are Stoker 1989 [1160], do Carmo 1976 [451], Berger and Gostiaux 1988 [175], among others. See also the new events in Arnol’d 1994 [65] and other references in §§1.4.6.

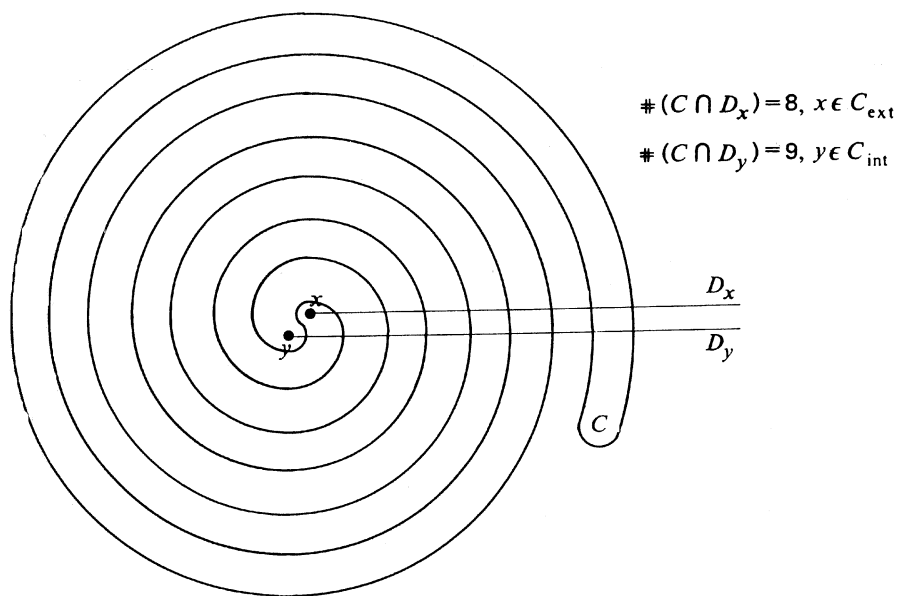


Fig. 1.21. A trick to find when a point belongs to the interior or to the exterior of a curve

Theorem 2 (Jordan curve theorem) *A simple closed plane curve has an interior and an exterior.*

This implies the more common statement that a simple plane closed curve separates the plane in two regions. More precisely the plane with the curve